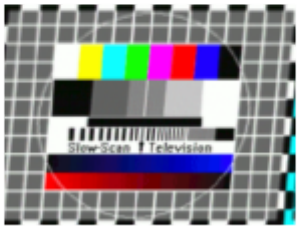
SSTV Encoding and Decoding + Image Processing

SSTV Function:

- Takes roughly ~3kHz bandwidth
- Image processing time ranges from 8s-2m+ based on mode/setting
- Utilizes analog frequency modulation to transmit brightness values of individual pixels
    - Frequency shifts up to indicate brighter pixel image, down for darker. Luminance is encoded on a scale from 1500 Hz (black) to 2300 Hz (white) with grey tones in between.
- Color is achieved by sending separate color codes individually, typically RGB values.
- “Sync is 1100 to 1300 Hz; channel information is 1500 to 2300 Hz.”
- Frame Sequential Transfer - Older form of image processing. Equal length frames of RGB color information are transmitted to be recomposited and processed by the receiver. Potentially problematic due to greater potential for interference or signal fading in between syncs. Example



“Progressive forward-slanting of received frame due to slower sampling clock rate at the receiver”



“Righthand frame depicts a reverse direction slanting due to a faster sampling clock.”

- Line Sequential Transfer - What most modern SSTV modes use. Process RGB color composition one scan line at a time in sequence. Less susceptible to corruption.
- Most SSTV modes involve synchronous detection, involving vertical interval signaling (VIS)[1], to synchronize between the transmitting and receiving ends of the circuit; a mismatch in the clock speed between the transmitter and receiver can lead to a tilt-like distortion to appear across the scanlines. So it’s important to make sure the transmitter (sound card DAC) clock and

[1] - Vertical Interval Signaling (VIS) Codes are used to vertically synchronize the scan lines of the picture. More information, including how they are encoded, can be found at https://www.digigrup.org/ccdd/sstv.htm

    - “Most SSTV software allows the operator to either perform corrections by introducing clock offset during the reception or to allow the user to do a post-reception slant correction”
    - This is only a problem with “free-run” SSTV such as Martin and Scottie SSTV mode. Asynchronous modes such as Robot do not experience this clock desynchronization; however, they have their own issues with interference and desyncing.
    - I’ve heard people use 11.025kHz as their sampling rate to mitigate this, but also 44.1 and 48kHz have been used so I need to read up more on this.
- There are many open source programs that transmit SSTV images for us in a variety of modes. A popular one is MMSTV

Modes in SSTV:

Types of SSTV Modes:

Robot



Robot 72 Image (~72s)

- “The Robot modes are named according to the converters in which they were implemented. They are scan-converters Robot 400C, 450C and 1200C. They were produced in San Diego by Robot Research Inc.”
- Do not use RGB values to encode color but rather YCrCy (Luminance, Chrominance R-Y, Chrominance B-Y) or YC (Luminance/Chrominance)
    - This is allegedly more efficient than RGB color encoding, meaning faster image processing
    - Luminance is encoded on a scale from 1500 Hz (black) to 2300 Hz (white) with grey tones in between.
    - This can be tricky; however, as the receiver needs to be perfectly tuned to the SSTV signal. Any interference will distort the color when the frequency received is off by more the ±50 Hz
        - To compensate, Robot Research introduced “gray gradation scaling.” Here the first 8 to 16 scanlines are purely reserved for the

gradation scale which provides a reference for the receiver to auto-tune the picture to. This header space is also sometimes used for storing info such as call signs or station data.

    - Luminance being encoded separately allows you to transmit and receive black and white images as well with little modifications which isn't compatible with RGB SSTV modes
    - https://pub.smpte.org/doc/st170/20041130-pub/st0170-2004_stable2010.pdf - This should show how to make the conversion from YC_rC_b to RGB if needed
- Robot 24, Robot 36, and Robot 72 are named after their image processing speeds. They receive and process images in 24, 36. And 72 seconds respectively
- The resolution of Robot SSTV are as follows:

Name	Lines
Robot 8	120[1]
Robot 12	128 luma, 32/32 chroma × 120[2]
Robot 24	128 luma, 64/64 chroma × 120
Robot 32	256 × 240[3]
Robot 36	256 luma, 64/64 chroma × 240
Robot 72	256 luma, 128/128 chroma × 240

- The robot system uses asynchronous transfer and needs a way to detect vertical sync.
    - "The VIS code is sent as: 30 mS start of 1200 Hz. 7 data bits sent LSB first, 30 mS each, 1100 Hz for 1, 1300 Hz for 0. Even parity bit of 30 mS. 30 mS stop of 1200 Hz."

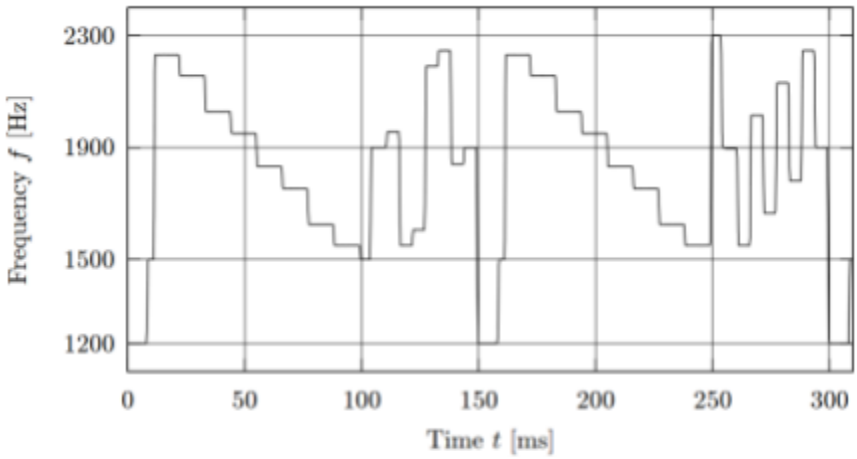
[1] - "You will notice that number of pixels per line is not listed. This is because SSTV uses analog modulation which can be sampled at any rate. Various systems might sample the same scan line with 256, 320, 512, 640, or other number of pixels. They are all compatible but those with more samples per scan line produce better detail." -https://www.digigrup.org/ccdd/sstv.htm

In these modes, the decoder spends a fixed amount of time on each line. The decoder then divides the time across the pixels to be sampled on a line. For example: Robot 8 = 8 seconds total; 66.7ms/line. If they are sampling 320 pixels/line, then each pixel would be processed in ~0.21ms. Note: This does not include delays for horizontal/vertical sync

[2] - This refers to the composition of the luminance/chrominance mentioned earlier. "256 luma, 128/128 chroma × 240" essentially means 256 pixels of the luminance samples, 128 R - Y pixels and 128 B - Y pixels of the chrominance samples, and 240 total scanlines.

[3] Robot 36 (along with Robot 8) uses RGB

    - "The original 8 second B&W SSTV mode (about 30 years ago) used 5 milliseconds of 1200 Hz for horizontal sync”
- “The scan-line is composed of the starting sync, followed by a short 3.0ms gap of 1500 Hz and then the image part with luminance and chrominance. The chrominance differential signals begin with additional sync pulses. The 1500Hz sync is before *R - Y* and the second 2300Hz is before *B - Y*. Due to the additional sync with a different sync frequency, it is possible to re-synchronize 4 : 2 : 0 formats after an interruption. The chrominance syncs are separated from the scan-line with 1500Hz gap that lasts 1.5 ms”



Two Scanlines of a Robot 36 Color Signal

https://www.sstv-handbook.com/

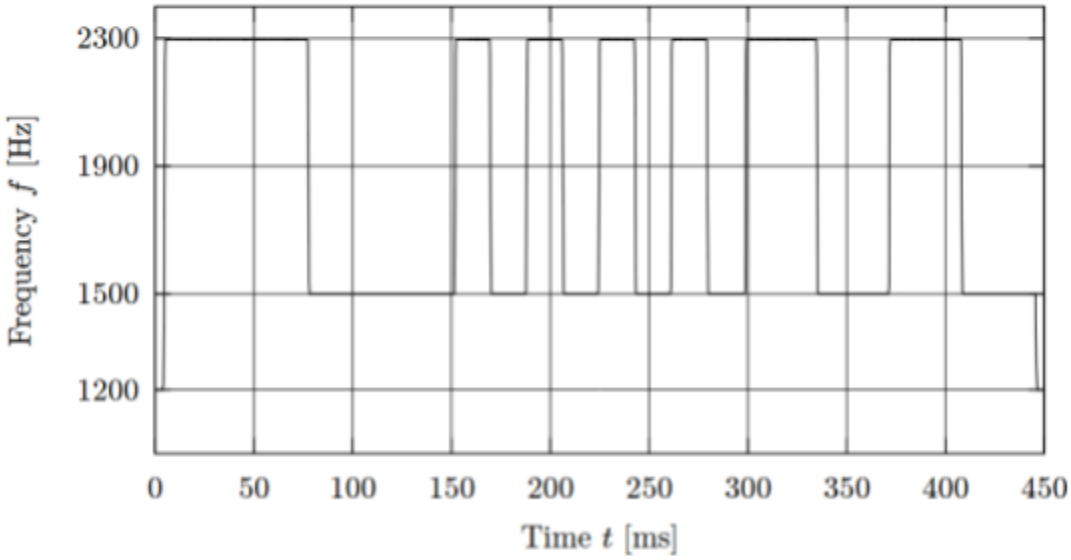
Martin

- The Martin SSTV mode was created by Martin Emmerson, G3OQD.
- Uses RGB values to store pixel data in the scan line
    - The colors are sent in the order green, blue, then red to adhere to the standard promoted by the WRASSE SC-1; however, changing the order of the hues does not affect image quality



Martin 1 Image (~60s)

- Martin SSTV involves a synchronous transfer system where each individual scan line is preceded by a VIS signal lasting about 4.862 ms.



Scan-line of Martin M1 when Color Bars are Sent

- This means, compared to the asynchronous transfer system that the Robot SSTV mode uses, only one sync signal is sent for each scan line
- Due to how the sync signals are processed, this method of processing the image signal will not degrade the color information.
- “In time intervals where the line sync is not transmitted, the gaps are filled with a reference level of black at 1500 Hz for 0.572ms”

Scottie

PD

Links:

- SSTV-Handbook
- PD Modes
- Transmission Modes of SSTV

Additional Readings I Found Helpful:

- Scan Line
- si4703
- KD2DB - Revisiting Slow-Scan Television
- KD2BD - The HRC SSTV Notebook
- QST SSTV Magazine
- British Amateur Television Club - SSTV Explained
- BruXy - Robot 1200C SSTV
- W5ZR - Slow Scan Television History
- VA4ROM, All Things Digital - SSTV Explained
- 101 Things - Slow Scan Television (SSTV) Decoder



Robot 36s Color



Martin M1